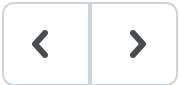




Course Schedule



SCHEDULE

ALL TIMES EASTERN - Please see the [Course and Program Policies](#) section of the Course Outline for details

Week / Module	Module Topics	Live Sessions	Readings, Activities and Assignments	End/Due Date	Weight (%)
Week 1 May 26-June 1 Module 1: Reliable and Scalable Data – Intensive Applications	• Course logistics and system setup • Issues of scaling to Big Data • Reactive Design framework • Strategies that enable high reliability, scalability, resilience	Week 1 Live Session: Thursday, May 29, 6:00 p.m. - 7:30 p.m.	Setup and Installation		
			Complete all of Module 1		
			Introduce Yourself	Sunday, June 1 at 11:55 p.m.	Ungraded
Week 2 June 2-8 Module 2: Relational Databases	• The relational model • SQL • Operational data stores and data warehouses	Week 2 Live Session: Thursday, June 5, 6:00 p.m. -	Complete all of Module 2		
			Review the Assignment 1: SQL Queries instructions		

		7:30 p.m.			
Content below will become available on Monday, June 9 at 9:00 a.m.					
Week 3 June 9-15 Module 3: NoSQL	- Key-value stores - Column-oriented databases - Document-oriented databases - Object, graph, and triple stores - MongoDB	Week 3 Live Session: Thursday, June 12, 6:00 p.m. - 7:30 p.m.	Check group membership for Group Assignment	Monday, June 9 at 9:00 a.m.	
			Complete all of Module 3		
			Submit Assignment 1	Sunday, June 15 at 11:55 p.m.	15%
			Review the Assignment 2: MongoDB instructions		
Week 4 June 16-22 Module 4: Distributed Datastores	- Distributed filesystems - The scalability problem for relational databases - The CAP theorem - Replication and partitioning - The Hadoop distributed filesystem	Week 4 Live Session: Thursday, June 19, 6:00 p.m. - 7:30 p.m.	Complete all of Module 4		
			Submit Assignment 2	Sunday, June 22 at 11:55 p.m.	15%
Week 5 June 23-29 Module 5: Introduction to Spark	- Setting up Spark - Spark datasets and dataframes	Week 5 Live Session: Thursday, June 26, 6:00 p.m. - 7:30 p.m.	Complete all of Module 5		
			Complete Early Course Evaluation Survey	Sunday, June 29 at 11:55 p.m.	Ungraded
			Review the Assignment 3: Spark instructions		

Week 6 June 30-July 6 Module 6: Analytics with Spark	- Spark actions - Spark transformations	Week 6 Live Session: Thursday, July 3, 6:00 p.m. - 7:30 p.m.	Complete all of Module 6		
			Review the Assignment 4: Spark ML instructions		
Week 7 July 7-13 Module 7: Spark Streaming	- What's different about streaming data - Stream processing in general - Spark stream processing	Week 7 Live Session: Thursday, July 10, 6:00 p.m. - 7:30 p.m.	Complete all of Module 7		
			Submit Assignment 3	Sunday, July 13 at 11:55 p.m.	15%
Week 8 July 14-20 Module 8: Data in Motion	- Meta-data - Patterns of data transfer - Formats for serializing and transmitting data	Week 8 Live Session: Thursday, July 17, 6:00 p.m. - 7:30 p.m.	Email instructor Group Assignment topic for approval	Monday, July 14 at 11:55 p.m.	
			Complete all of Module 8		
			Submit Assignment 4	Sunday, July 20 at 11:55 p.m.	15%
Week 9 July 21-27 Module 9: Big Data Architecture	- Lambda architecture - Lakehouses - Kafka and realtime analytics	Week 9 Live Session: Thursday, July 24, 6:00 p.m. - 7:30 p.m.	Complete all of Module 9		
			Group Assignment Discussion (for Part 2) available	Discussion Available: Sunday, July 27 at 11:55 p.m.	
Week 10 July 28-August 3	- Amazon - Azure - Google	Week 10 Live Session:	Complete all of Module 10		

Module 10: Cloud Analytics		Session: Thursday, July 31, 6:00 p.m. - 7:30 p.m.			
Week 11 August 4-10 Module 11: MLOps	• Introduction to MLOps • Continuous deployment • Tools for MLOps	Week 11 Live Session: Thursday, August 7, 6:00 p.m. - 7:30 p.m.	Complete all of Module 11		



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Activity Details

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